POLITECNICO MILANO 1863 | SCUOLA DI INGEGNERIA INDUSTRIALE E DELL'INFORMAZIONE

Adaptive Tuned Mass Damper

Course: Experimental Dynamics and Data analysis

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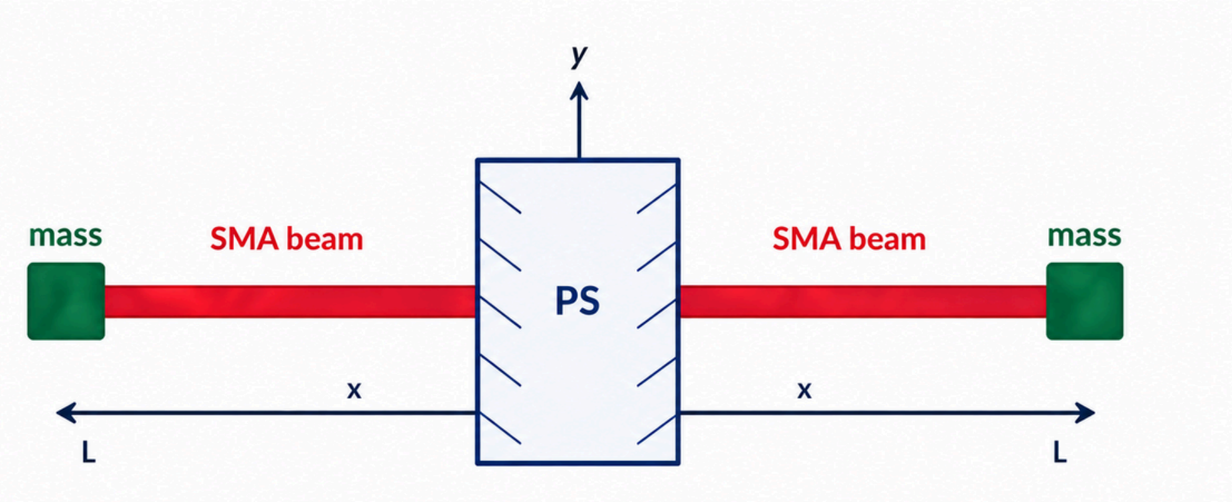
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Introduction & Experimental Setup

01

Introduction



Topic: analysis of first two modes of an Adaptive Tuned Mass Dampers made

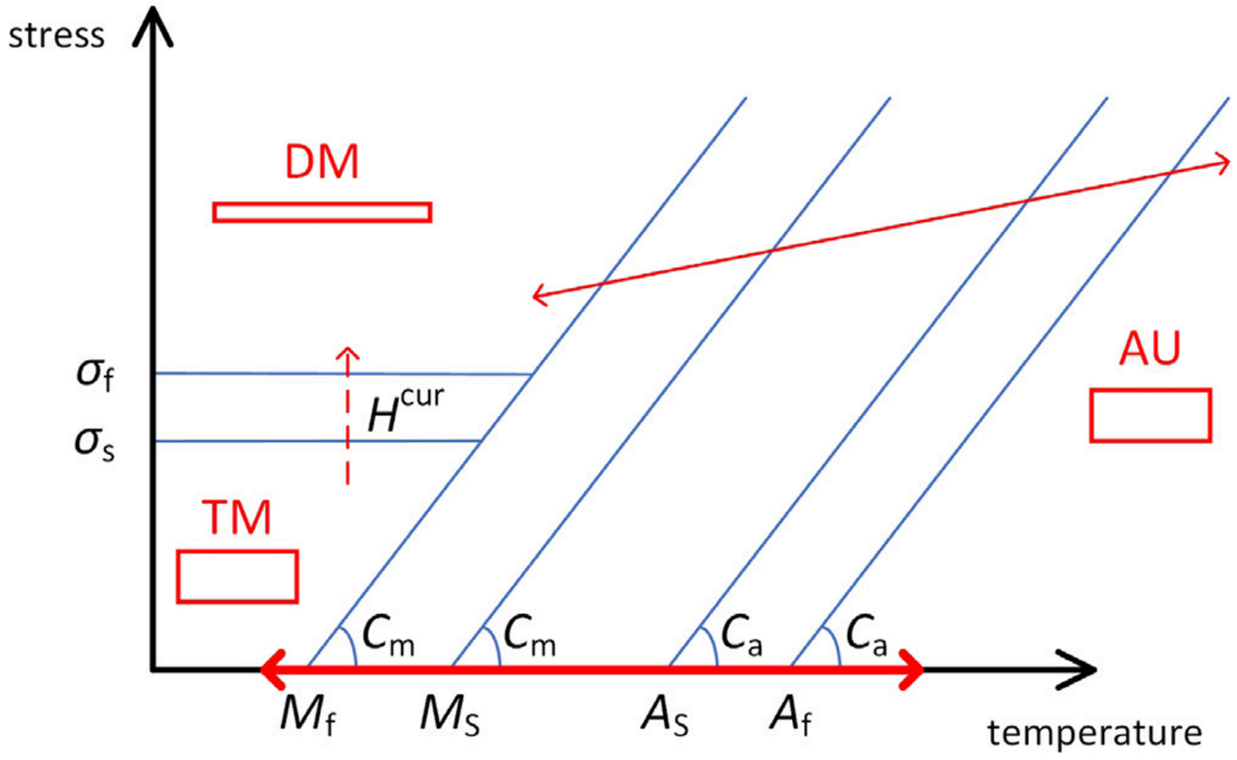
Request: extracting modal parameters of the structure and plotting eigen-frequencies and damping ratio as function of current

Modus operandis:

- full understanding of paper
- test in laboratory
- plotting of acquired data
- analysis and observation of obtained results

Assumption:

- Only heating is considered
- Transformation from Martensite to Austenite driven exclusively by temperature changes
- Mathematical treatment considers only small amplitude oscillation
- Focus only on the first two modes
- Mass at boundaries considered as a concentrated mass with translational and rotational inertia



Material model

- Martensite percentage coefficient:

$$\xi = \begin{cases} 1, & T \leq A_s^\sigma \\ \dfrac{A_f^\sigma - T}{A_f - A_s}, & A_s^\sigma < T < A_f^\sigma \\ 0, & T \geq A_f^\sigma \end{cases}$$

- Young modulus:

$$E = E_a + \xi \left(E_m - E_a \right) \longrightarrow \text{linear relation}$$

- Thermal coefficient:

$$\alpha = \text{constant}$$

Thermal model

- Power balance: $A_{\text{ext}}\, h\,(T - T_0) = Ri^2$
- Resistance: $R = \dfrac{\rho L_{\text{tot}}}{A}$ ⟶ $\rho = \rho_a + \xi\,(\rho_m - \rho_a)$ ⟶ linear relation
- Convective heat transfer coefficient: $h = \dfrac{N_u \psi_A}{D}$

  - Nusselt number $N_u = 0.3 + \dfrac{0.62 R_e^{1/2} P_r^{1/3}}{\left[1 + \left(\frac{0.4}{P_r}\right)^{2/3}\right]^{1/4}} \left[1 + \left(\dfrac{R_e}{282000}\right)^{5/8}\right]^{4/5}$
  - Reynolds number $R_e = \dfrac{r_A v_c D}{b_A}$
  - Prandtl number $P_r = \dfrac{\eta_A b_A}{\psi_A}$

with ν_c the velocity of the cylinder in vertical direction

Analytical model (1)

$$\gamma^4 = \frac{m\omega^2}{EJ}$$

Equation of motion of a beam:

$$w(x,t) = [B_1 \sin(\gamma x) + B_2 \cos(\gamma x) + B_3 \sinh(\gamma x) + B_4 \cosh(\gamma x)] \cos(\omega t + \phi)$$

Boundary conditions applied:

$$x = 0 \quad w(x,t) = 0$$

$$\frac{\partial w(x,t)}{\partial x} = 0$$

$$x = L \quad T(x,t) + M_a \frac{\partial^2 w(x,t)}{\partial t^2} = 0$$

$$M(x,t) + I_G \frac{\partial^3 w(L,t)}{\partial x\, \partial t^2} = 0$$

Frequencies evaluation:

$$[H]\vec{x} = \vec{0} \longrightarrow \det([H]) = 0$$

Parameters evaluation:

$$[H(\omega_j)]\vec{x} = \vec{0}$$

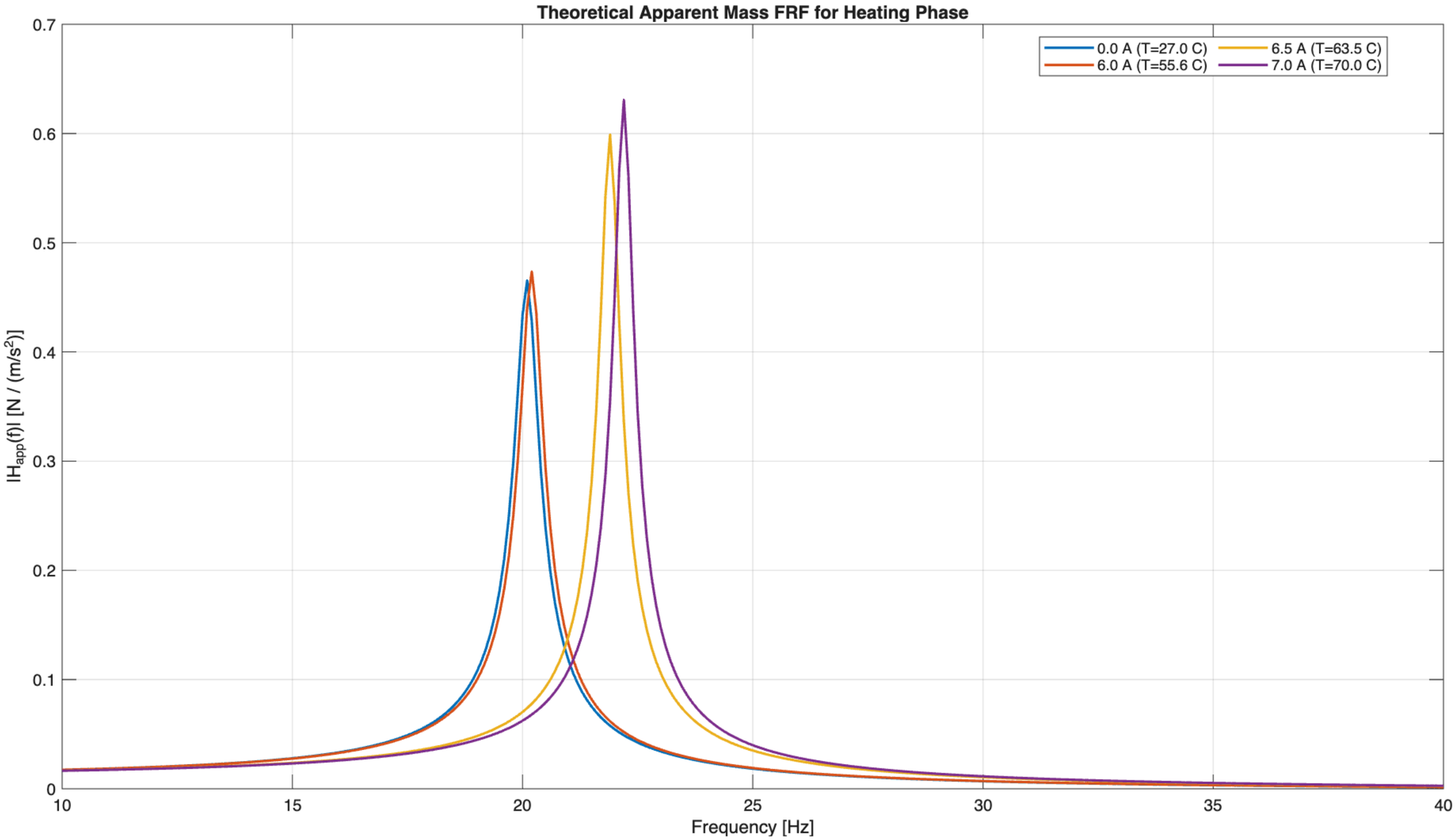
Analytical model (2)

Built the FRF:

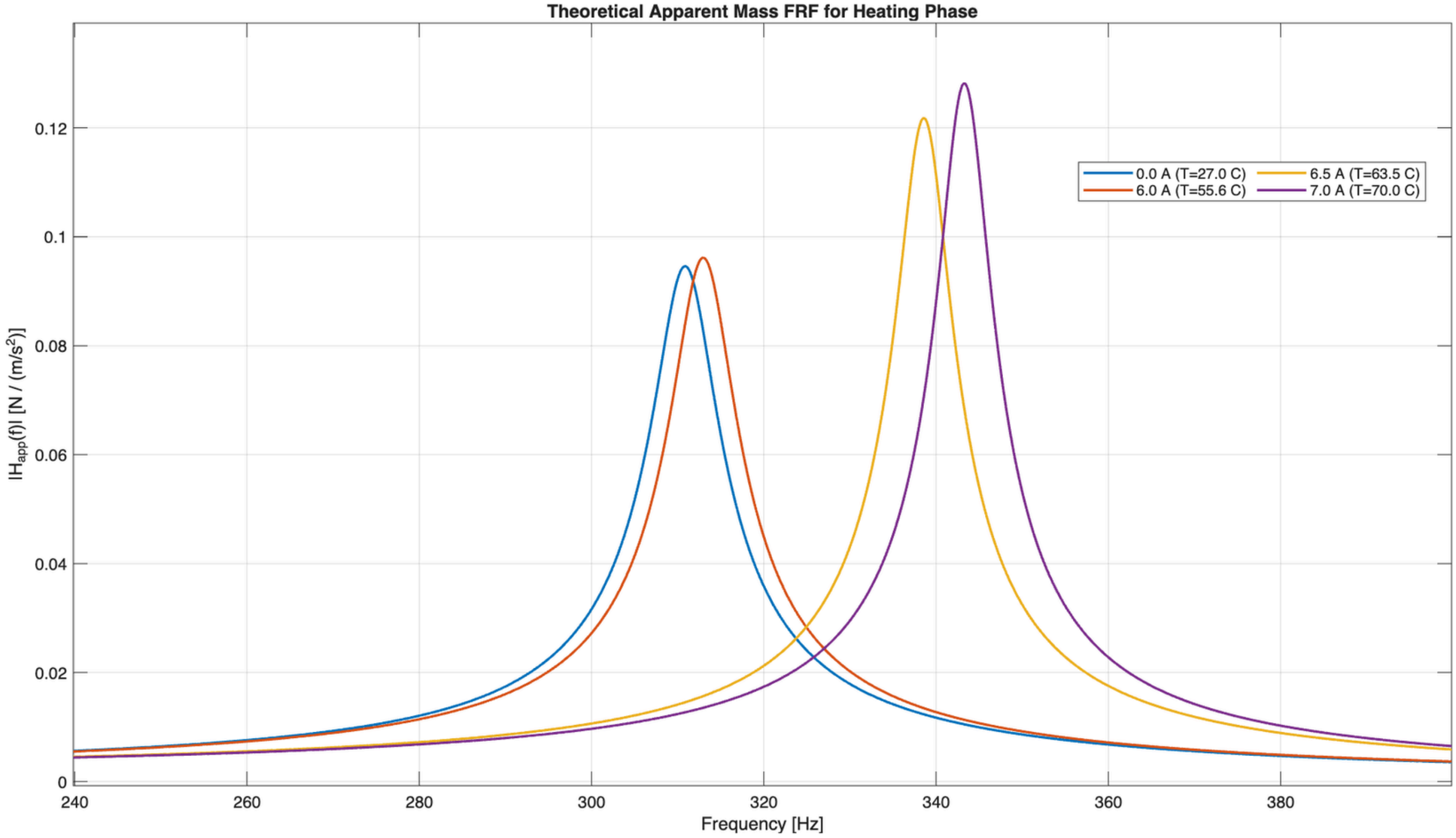
- S = vertical force of the beam that acts on the oscillator
- Y = vertical displacement of the oscillator

$$G_{SY}(j\Omega) = \frac{S(j\Omega)}{Y(j\Omega)} = -2EJ \sum_{i=1}^{n} \frac{\Omega^2 \phi_i'''(0) \left[\rho_m A \int_0^L \phi_i(x)\, dx + M_a \phi_i(L)\right]}{m_i \left(-\Omega^2 + 2j\xi_i \omega_i \Omega + \omega_i^2\right)}$$

Analytical model: first mode



Analytical model: second mode



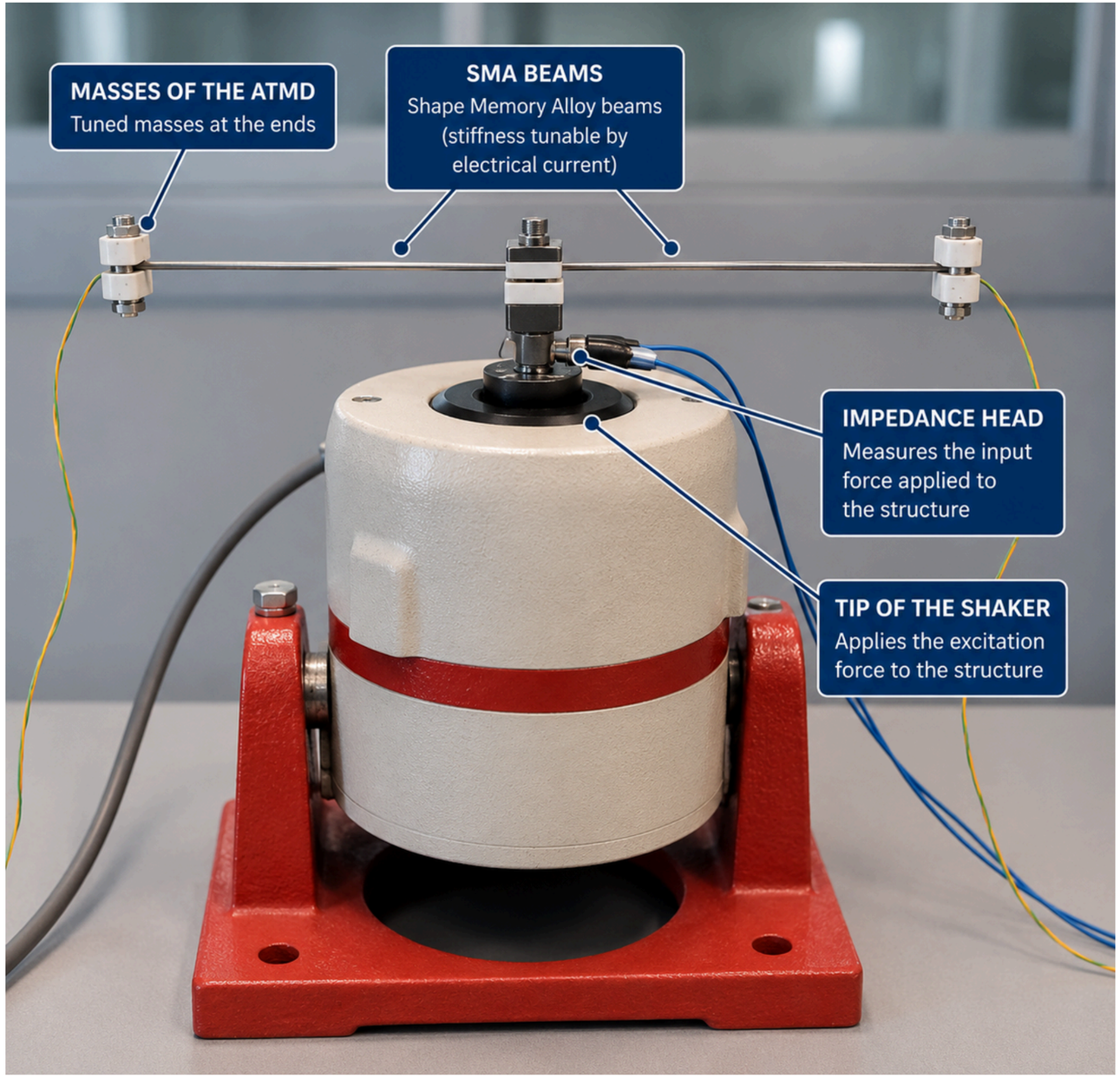
Experimental setup

Test specimen:

- beam made of SMA
- Tip mass attached to the free end
- Electrical current used to change SMA stiffness

Instrumentation:

- Electrodynamic shaker (excitation)
- Impedance head (input force)
- Current power supply



Test configuration

01

Frequency range: 10–400 Hz

- Expected first resonance around 18–20 Hz and second one around 300 (reference paper).
- Upper limit extended to 400 Hz to include higher vibration modes.

02

Sampling frequency: 2000 Hz

- Minimum sampling rate allowed by the acquisition system.
- Satisfies the Nyquist criterion for the selected frequency range.
- Provides sufficient frequency resolution for modal identification.

03

Excitation Signal
Broadband random excitation

- Simultaneous excitation of multiple vibration modes.
- Suitable for Frequency Response Function estimation.

04

Current Levels: 0 A – 9.5 A

- Measurements repeated at different electrical current levels.
- Heating the SMA beams modifies their stiffness and therefore the modal properties.

Laboratory Procedure



Set current
Apply the selected current to the SMA beam.



Thermal stabilization
Wait until the temperature reaches steady state.



Random excitation
Excite the structure using broadband random noise.



Acquisition
Measure force and acceleration simultaneously.



FRF estimation
Compute the Frequency Response Function.

Methodology & Data Analysis

02

Signal Processing & FRF Estimation

Signal Processing Chain

01. Raw Voltage Signals (V)

Initial input data collection

02. Calibration to EU

Conversion using sensors sensitivities from data sheet

03. Hanning Window & 60% Overlap

Signal conditioning for FFT

04. Spectral Averaging

Smoothing random input excitation

05. Estimator (Apparent Mass)

Final FRF calculation

Data Conditioning & Estimation Details

Leakage Minimization: The Hanning window forces the signal to zero at boundaries, preventing discontinuity.

SMA Linearization: Spectral averaging of continuous random excitation mathematically smooths the SMA's inherent hysteresis and internal friction, yielding an equivalent linear FRF.

Output Noise Rejection: The H1 formulation assumes a noise-free input, effectively treating the SMA's non-linear harmonic distortions as uncorrelated output noise and averaging them out.

$$H_1 = \frac{G_{AF}}{G_{AA}}$$

Signal Processing & FRF Estimation

Coherence Validation

The coherence function (γ^2) was computed as a quality indicator of the experimental data, providing a frequency-dependent measure of the linear correlation between input force and structural response.

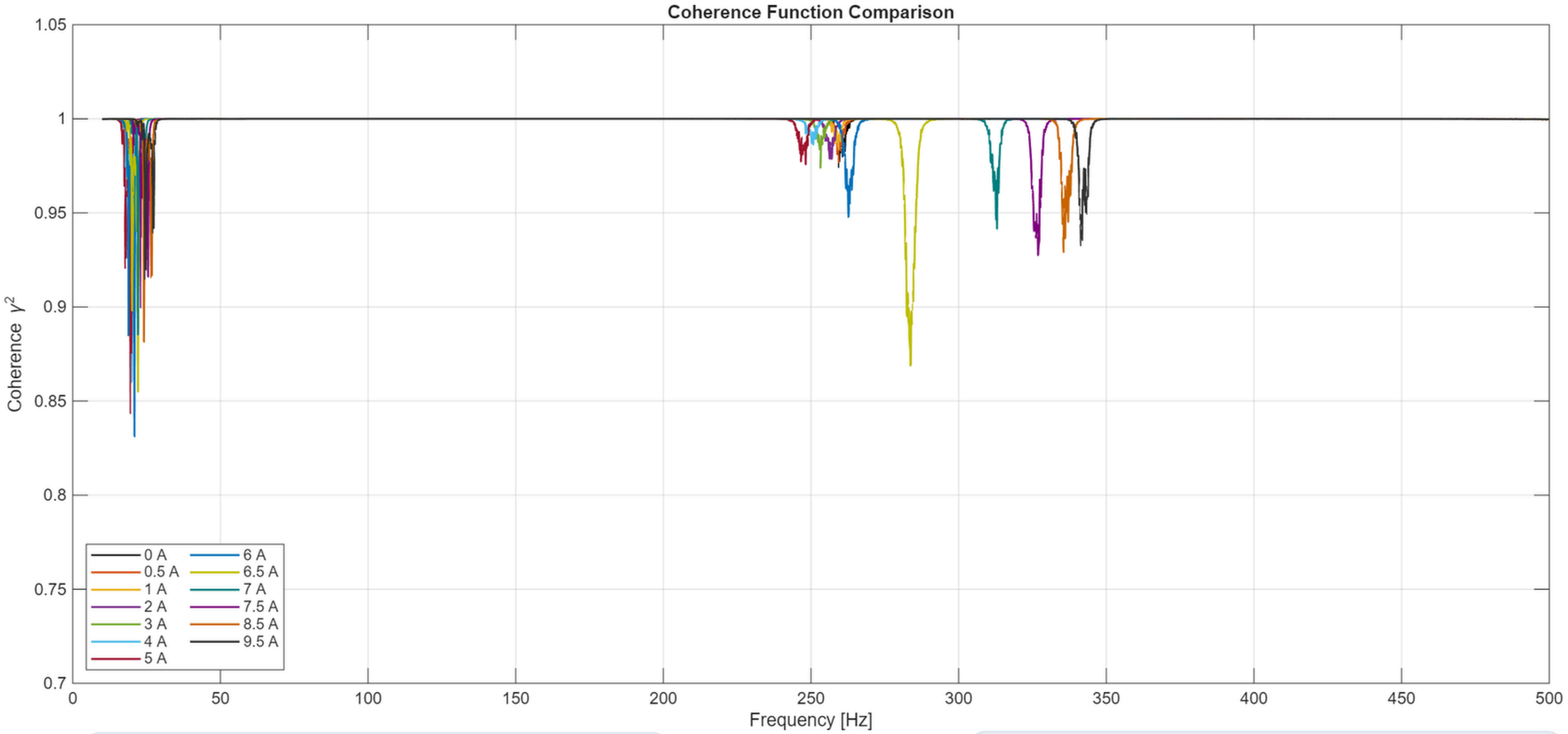
$$\gamma^2(f) = \frac{|G_{AF}(f)|^2}{G_{AA}(f).G_{FF}(f)}$$

Mass Cancellation

To isolate the ATMD dynamic response, the known mass of the clamping system (0.160 kg) was subtracted in the **frequency domain** to remove its integration effect.

$$H_{True} = H_{measured} + H_{clamp}$$

Coherence Function – Data Quality Validation



01

Data Quality

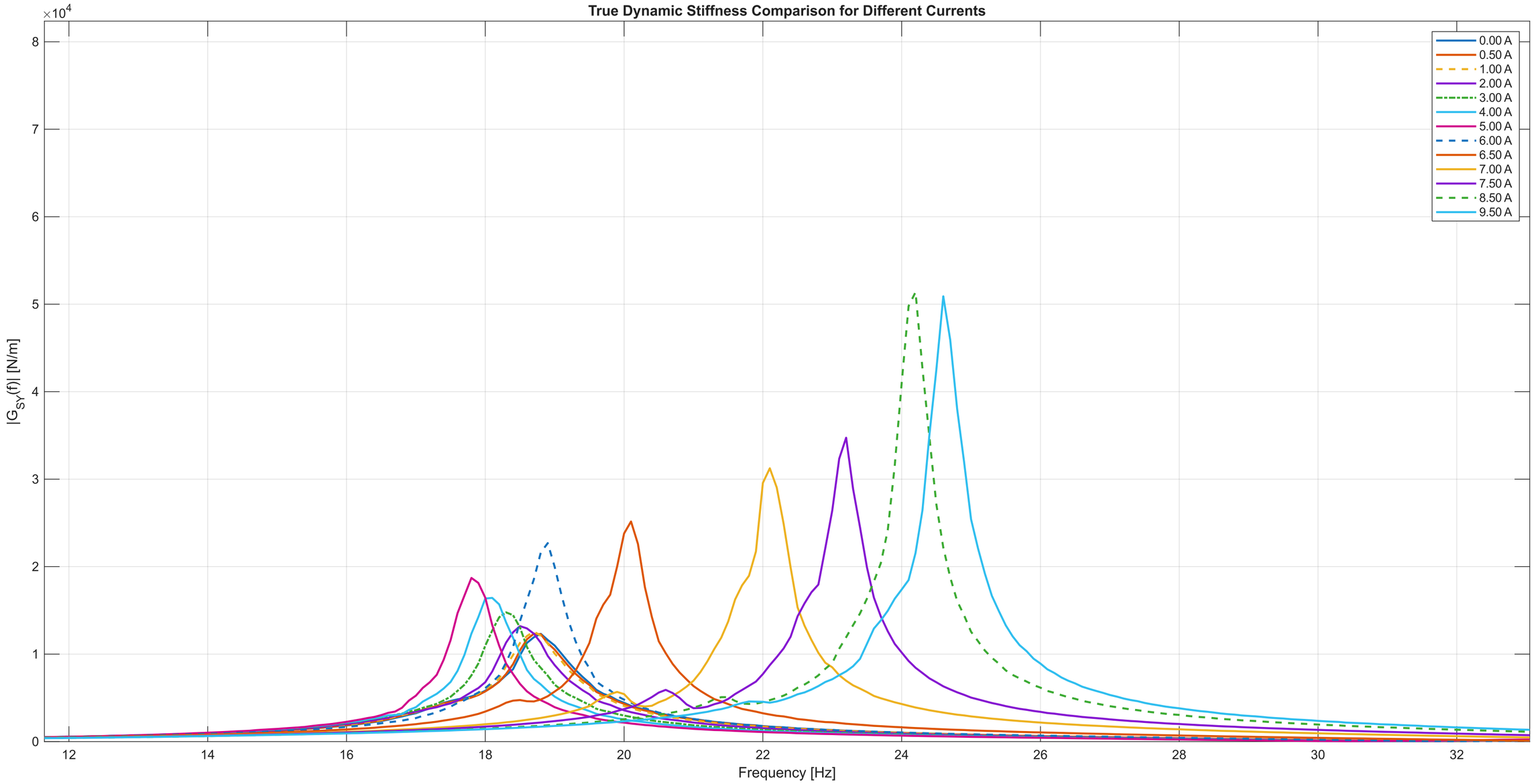
The coherence function remains above 0.95 across most of the 10-500 Hz range, indicating high-quality FRF measurements suitable for the H1 estimator.

02

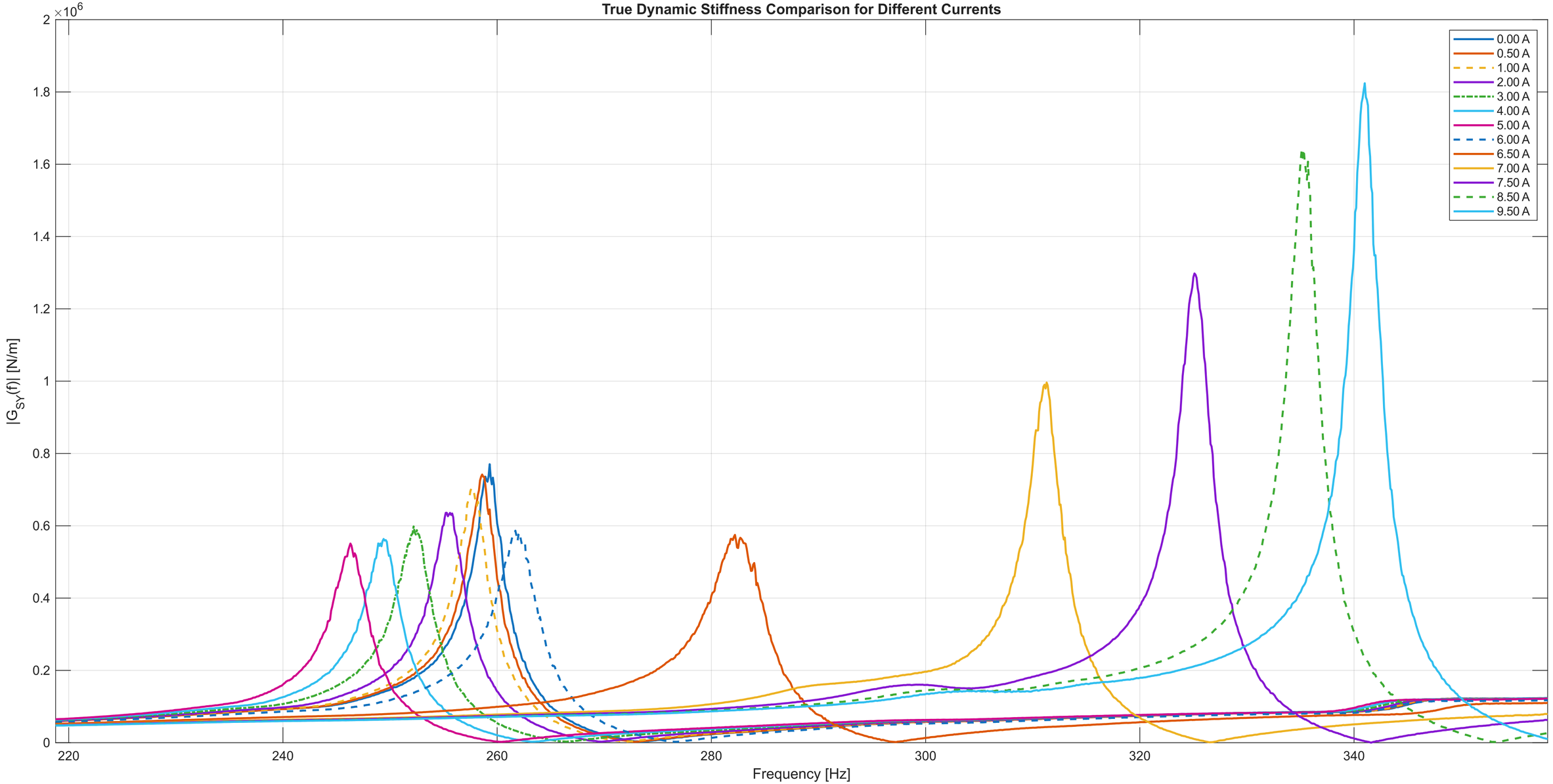
Performance at Peaks

Coherence through all the frequency range is above 0,84.
Which is very important for an accurate modal parameter extraction.

Experimental FRF for the first mode shape



Experimental FRF for the second mode shape



Modal Parameter Extraction Strategy

Step 1: Initial Estimation

Peak Picking

Half-Power Bandwidth (–3 dB) method applied to the FRF magnitude for preliminary identification of damping ratio.

Limitation

Struggles with spectral distortion and double peaks in non-linear SMA phase transitions, yielding unreliable damping estimates.

Step 2: Optimization

SDOF Least-Squares Minimization

- Advanced non-linear least-squares curve-fitting algorithm (MATLAB lsqnonlin).
- Iterative error minimization between measured FRF and parametric SDOF model.

Dynamic Windowing

Window constrained to **±10%** of resonance peak to ensure local convergence and reject baseline noise. The windowing also ensures SDOF local validity.

Modal Parameter Extraction Strategy

6-Parameter SDOF Analytical Model

Simultaneous error minimization across real and imaginary components of the complex-valued FRF.

$$H(\omega) = (C_{real} + i.\,C_{img}) + \frac{(A_{real} + i.\,A_{img})}{(\omega_n^2 - \omega^2 + i.2\zeta\omega_n\omega)}$$

Complex Modal Residue

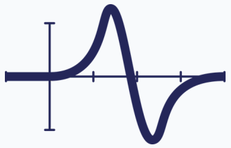
(A_{real}, A_{img})

Represents the complex modal contribution of the identified mode, accounting for both amplitude and phase of the FRF.

Out of Band Residuals

(C_{real}, C_{img})

Represents the contribution of modes outside the elected frequency band, approximated as a constant residual term over the fitting range to improve local FRF reconstruction.



Core Parameters

(ω_n, ζ)

Represents the physical modal parameters of the system: natural frequency and viscous damping ratio identified at each current level.

Theoretical Consistency of Optimization Results

The Optimization results for mode 1 current = 9.5 :

- Modal Residue: $A_{real} = -920.25, A_{img} = 22.35$
- Out-of-Band Residuals: $C_{real} = -0.0228, C_{img} = 0.0172$

- **Dominant Real Modal Residue:** The ratio of the imaginary to real modal residue is exceedingly small.
- **Proof of Proportional Damping:** Theoretically, a perfectly classically damped structure, yields a purely real residue. The near-zero imaginary component physically validates our SDOF assumption and proves the phase transition does not introduce heavy complex modal coupling.
- **Negligible Out-of-Band Influence**: The Out-of-Band residuals are on the order of 0.01, orders of magnitude smaller than the resonance peak. This confirms that higher-order modes (like Mode 2) and clamping mass effects are successfully isolated from the fundamental bending mode

Nyquist Plot Validation of the SDOF Assumption

The Concept

- **The Complex Plane:** The Nyquist plot graphs the Imaginary vs. Real components of the Frequency Response Function (FRF).
- **The SDOF Signature:** For an isolated resonance, an ideal Single Degree of Freedom system always maps to a perfect circular trajectory.

$$H(\omega) = \frac{1}{k - m\omega^2 + ic\omega}$$

Validation Methodology

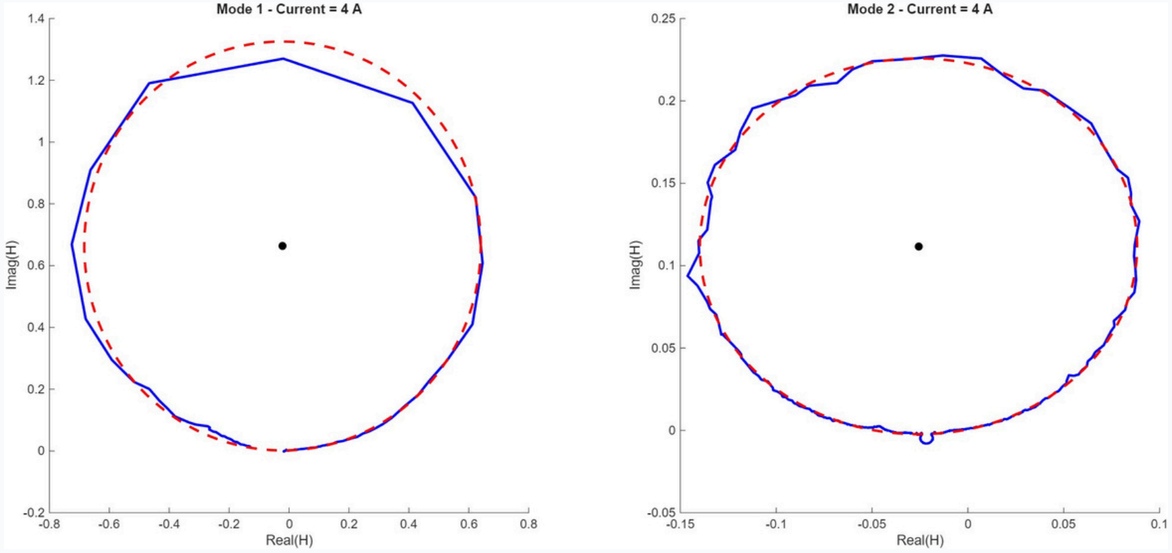
- **Geometric Fitting:** A circular curve is mathematically fitted to the experimental modal data points within the resonance window.
- **Statistical Verification:** The Single Degree of Freedom assumption is rigorously validated by minimizing the Root Mean Square Error (RMSE) and maximizing the Normalized Fit Quality (NFQ).

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(d_i - R)^2}$$

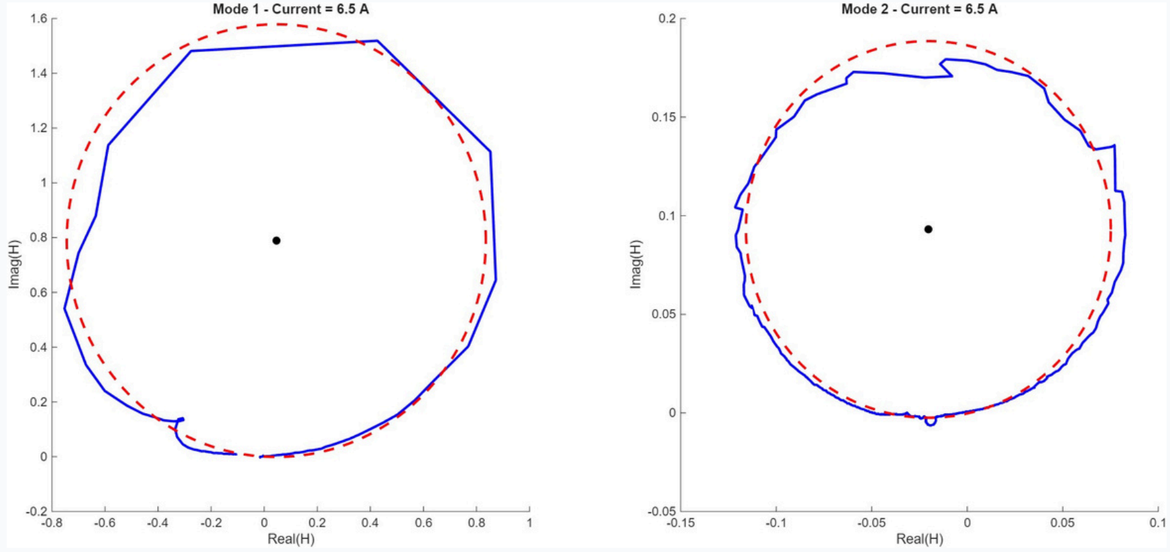
$$\mathrm{NFQ} = 1 - \frac{\sum_{i=1}^{N}(d_i - R)^2}{\sum_{i=1}^{N} d_i^2}$$

Nyquist Plot Validation of the SDOF Assumption

Real(H) - Imag(H) plot and fitted circle - Current = 4A



Real(H) - Imag(H) plot and fitted circle - Current = 6.5A



Nyquist Plot Validation of the SDOF Assumption

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(d_i - R)^2} \qquad \mathrm{NFQ} = 1 - \frac{\sum_{i=1}^{N}(d_i - R)^2}{\sum_{i=1}^{N} d_i^2}$$

SDOF assumption is validated

Very small RMSE and NFQ value close to 1 shows that the modal data points are almost circular, thus SDOF assumption is validated.

current(A)	RMSE_Mode1	NFQ_Mode1	RMSE_Mode2	NFQ_Mode2
0	3.19e-02	0.9959	2.56e-3	0.9989
0.5	3.52e-02	0.9950	2.24e-03	0.9992
1	3.03e-02	0.9964	2.08e-03	0.9993
2	2.89e-02	0.9969	1.99e-03	0.9993
3	2.46e-02	0.9981	1.98e-03	0.9992
4	2.37e-02	0.9985	1.68e-03	0.9994
5	2.90e-02	0.9982	2.09e-03	0.9991
6	3.56e-02	0.9975	2.76e-03	0.9984
6.5	5.25e-02	0.9939	3.60e-03	0.9968
7	6.66e-02	0.9898	4.76e-03	0.9954
7.5	5.96e-02	0.9913	3.92e-03	0.9978
8.5	4.51e-02	0.9968	3.70e-03	0.9987
9.5	5.52e-02	0.9949	4.13e-03	0.9987

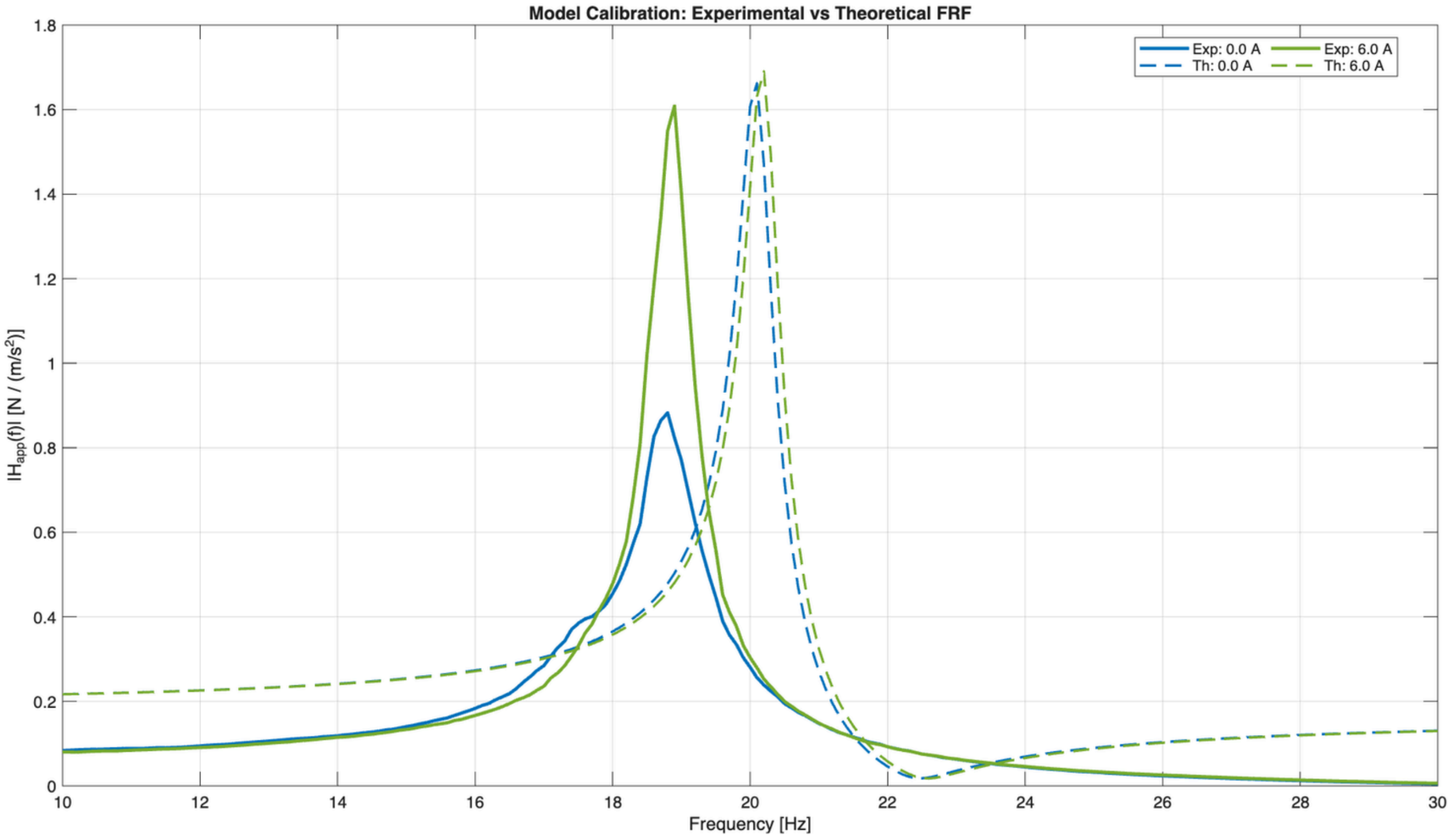
Results & Discussion

03

Results & Discussion

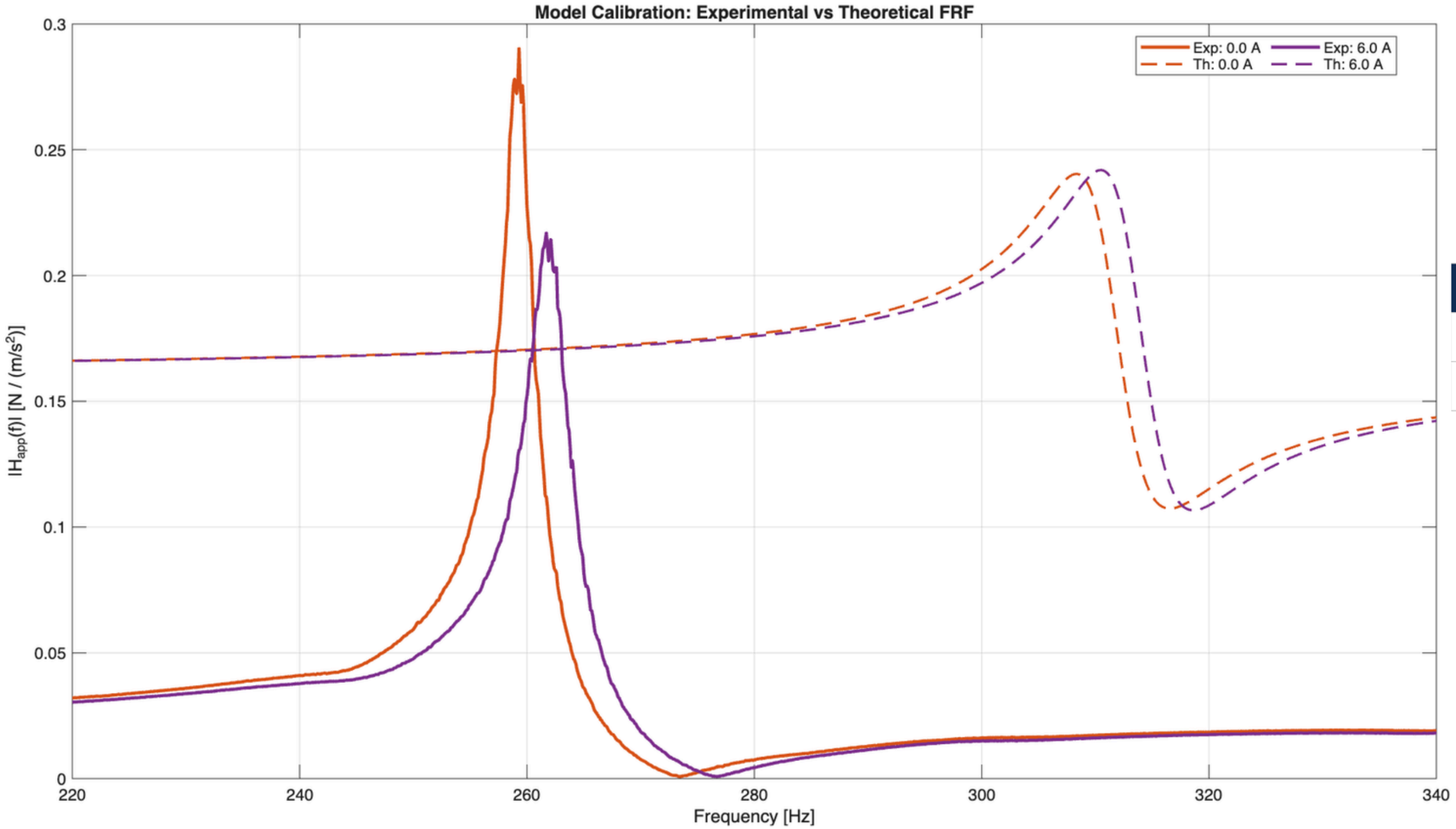


Experimental vs Analytical FRF for Mode 1



Current	Analytical	Experimental	ΔF
0 A	20.1	18.8 Hz	1.3 Hz
6 A	20.2 Hz	18.9 Hz	1.3 Hz

Experimental vs Theoretical FRF for Mode 2



Current	Analytical	Experimental	ΔF
0 A	308.3	259.3 Hz	49 Hz
6A	310.4 Hz	261.7 Hz	48.7 Hz

How to improve the analytical model: Frequency offset problem

- **Pre-Transition Thermal Softening:** The theoretical piecewise model assumes a constant Martensite stiffness (E = Em) for all temperatures below As. It fundamentally ignores the macroscopic thermal softening that lowers the physical natural frequency prior to phase transformation.

- **Idealized Clamped Boundary:** The Euler-Bernoulli mathematical matrix relies on a perfectly rigid wall. The physical laboratory setup contains microscopic mechanical compliance at the clamp.

- **Unknown parameters:** The length of the beam used in the test was not measured so the length mentioned on the paper was used instead, also the dimensions for the tip mass were not measured, hence the moment of inertia in the boundary conditions are approximations.

How to improve the analytical model: Shape of the peak problem

- **Multi-Axis Coupling & Asymmetry (Double Peaks):** The theoretical model assumes a perfectly straight, symmetrical beam vibrating exclusively in a single transversal plane. In reality, the tip mass causes an initial static bending. This static curvature, combined with microscopic clamping asymmetries, causes orthogonal bending and torsional modes to couple with the primary bending axis, creating the overlapping "double peaks" observed in the experimental FRFs.

- **Linear Damping Interpolation:** The theoretical model applies a simplified linear interpolation between ζ_m and ζ_a. This mathematical approach misses the complex, mixed-phase boundary friction that causes severe damping spikes during the physical transition (particularly in Mode 2).

How to improve the analytical model

Parameter	Analytical	Derivation	Improvements
Beam Lenght	140 mm	Paper	Measure the beam
Ea	32.1 GPa	Paper	-
Em	39.5 GPa	Paper	-
Intertia Ig	0.7e-6 kg*m^2	Estimated	Measure the mass dimension

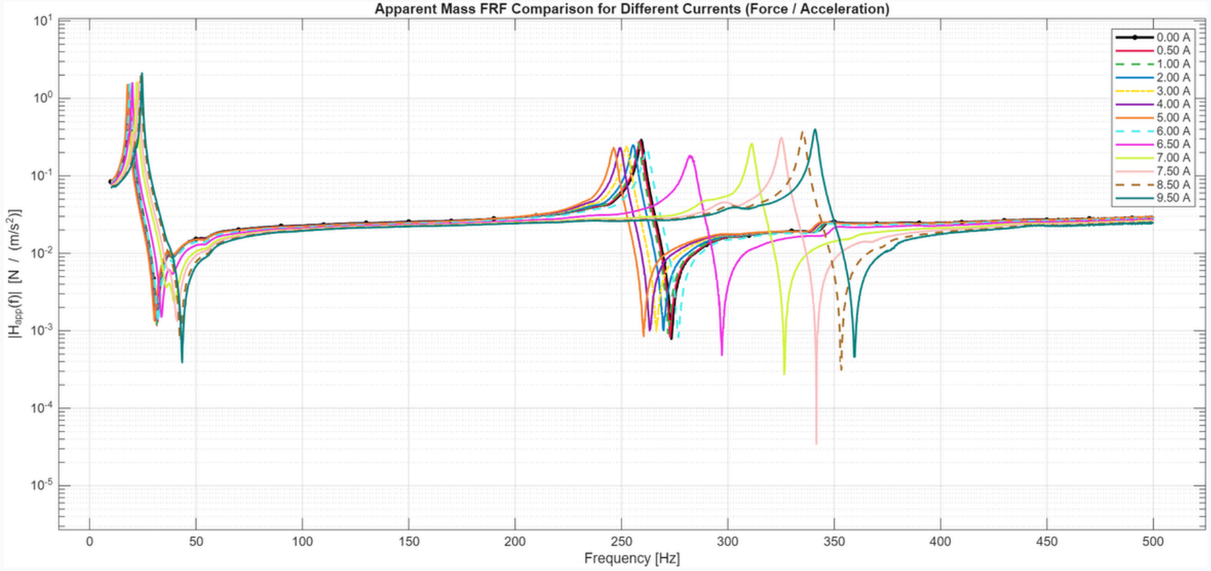
FRF Evaluation

The frequency response function (FRF) is measured between input base acceleration and output force from the SMA element. Two derived quantities are:

01

Apparent Mass

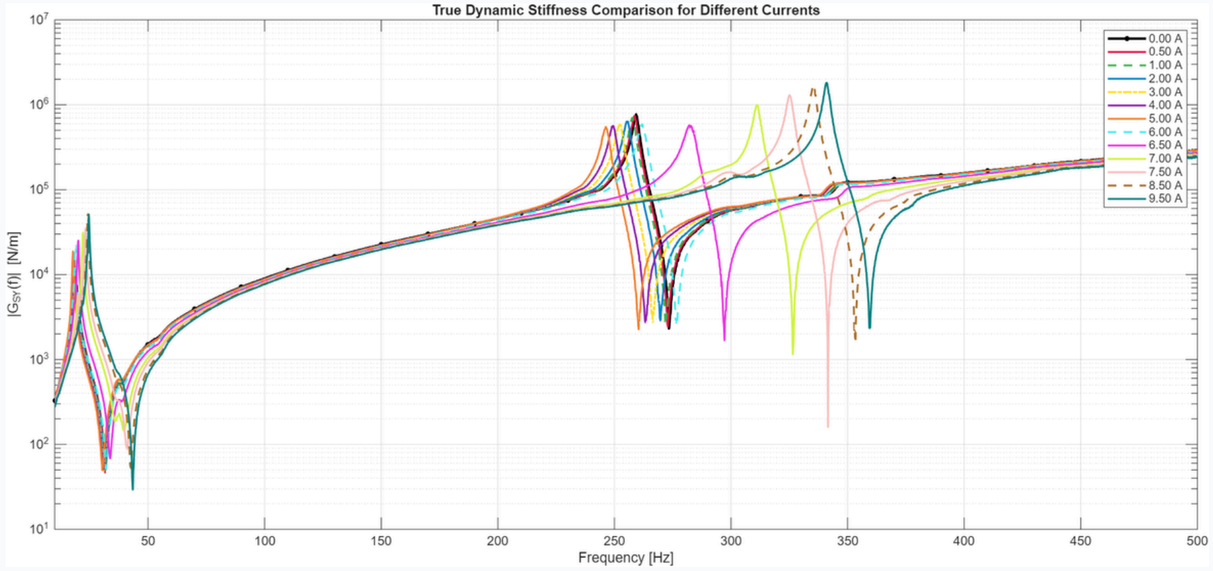
FRF expressed as Force/Acceleration (N/(m/s^2)), computed using the H1 estimator.



02

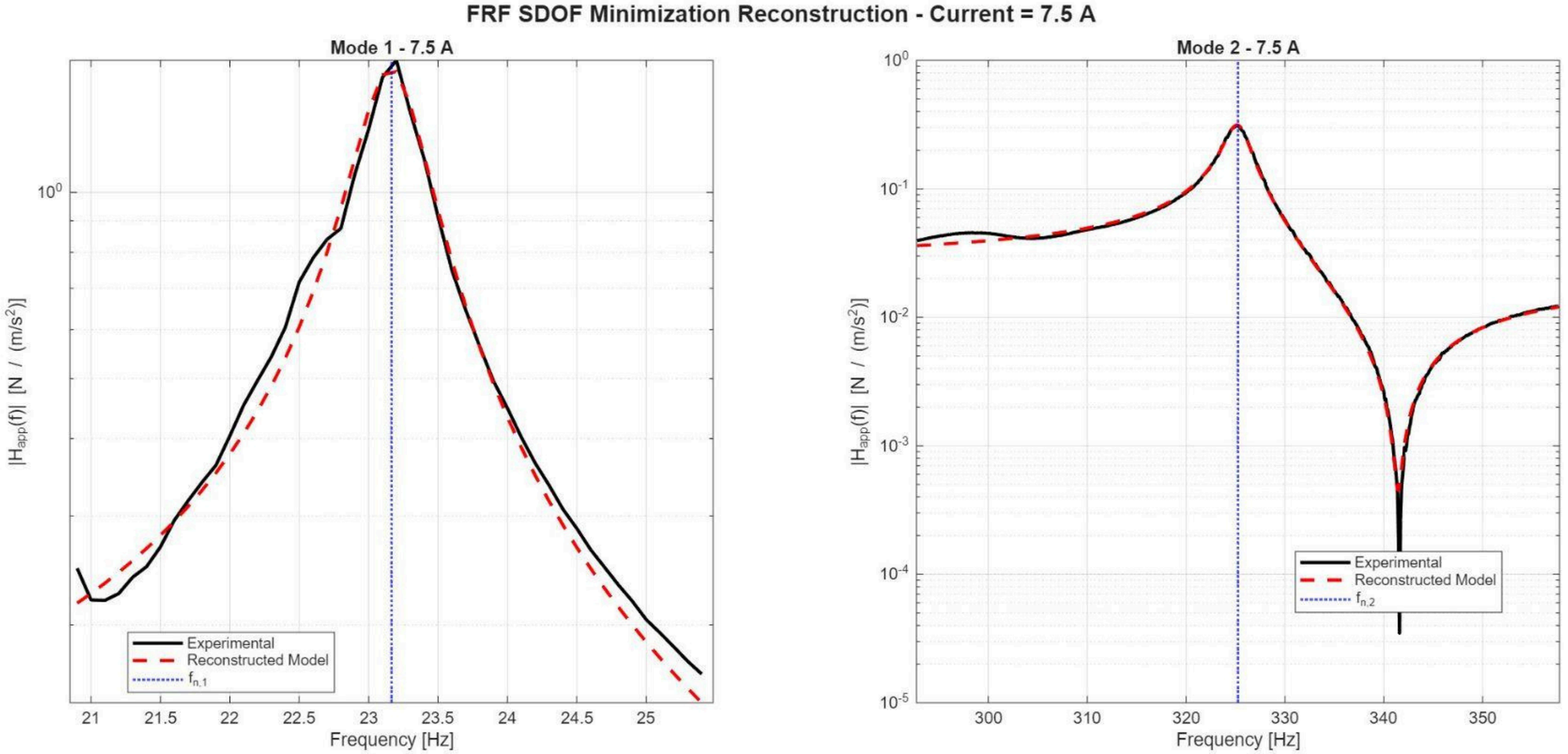
Dynamic Stiffness (G_SY)

Apparent Mass multiplied by –ω^2 , yielding frequency-dependent stiffness (N/m).



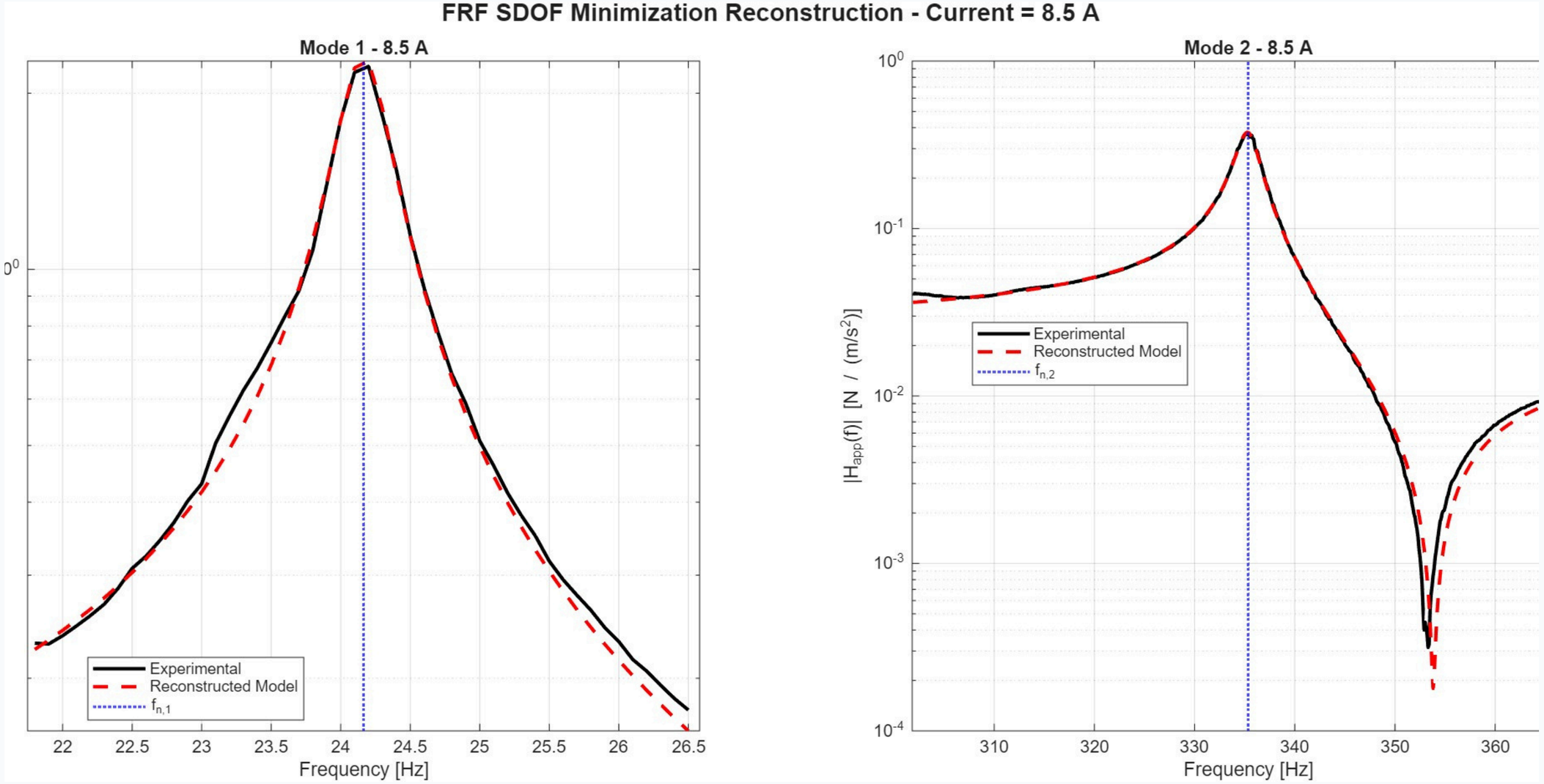
Modal Parameter Extraction & Validation

FRF Reconstruction — Current = 7.5 A

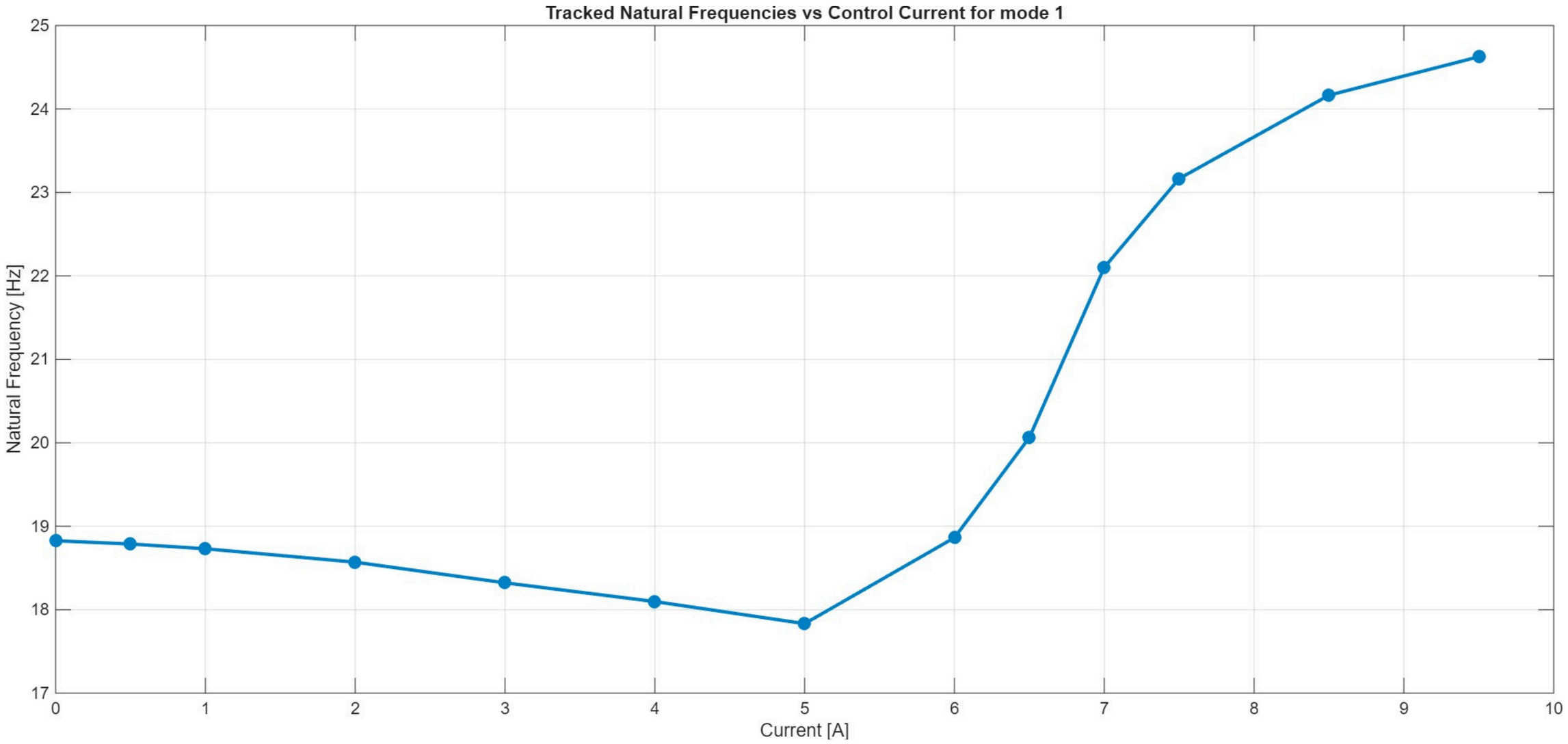


Modal Parameter Extraction & Validation

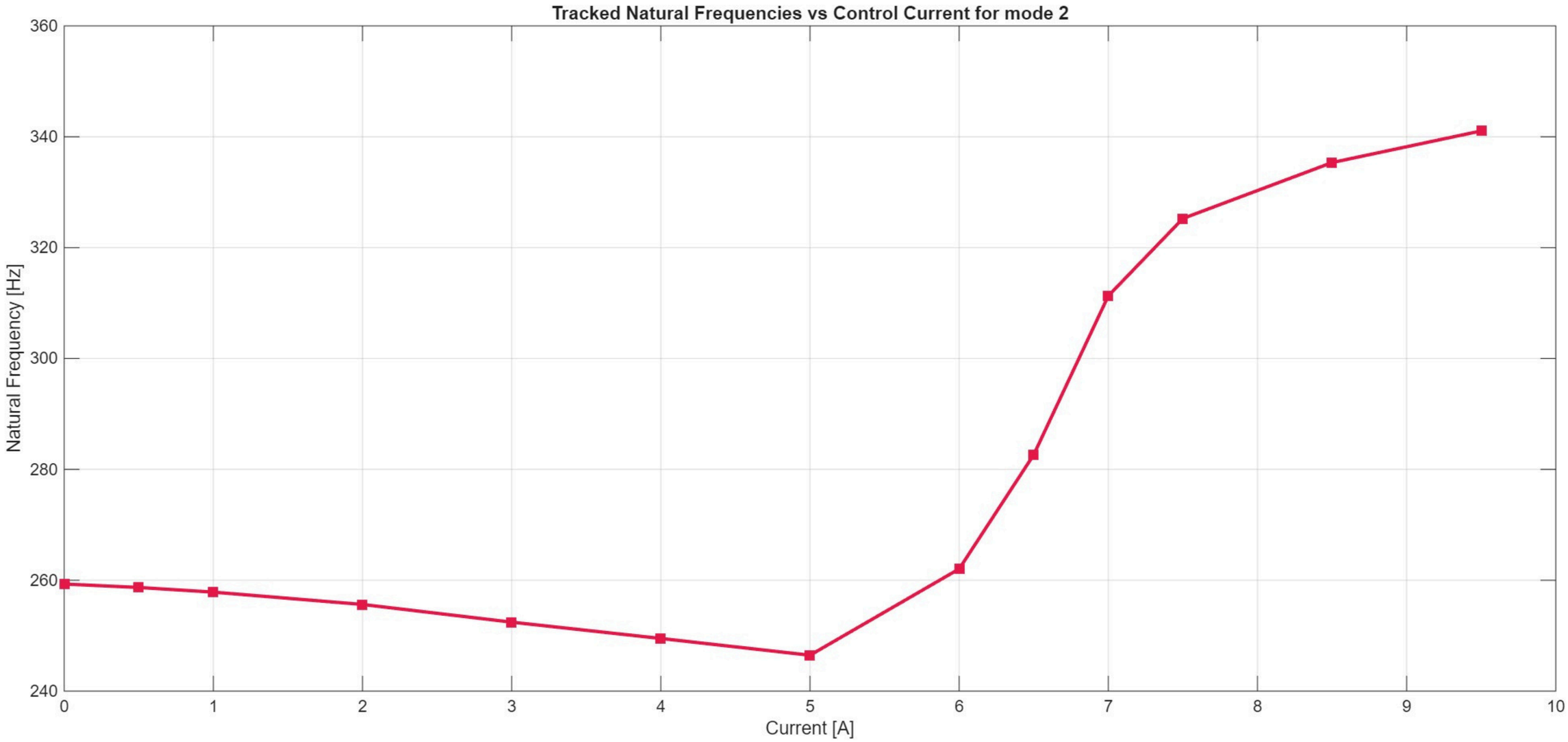
FRF Reconstruction — Current = 8.5 A



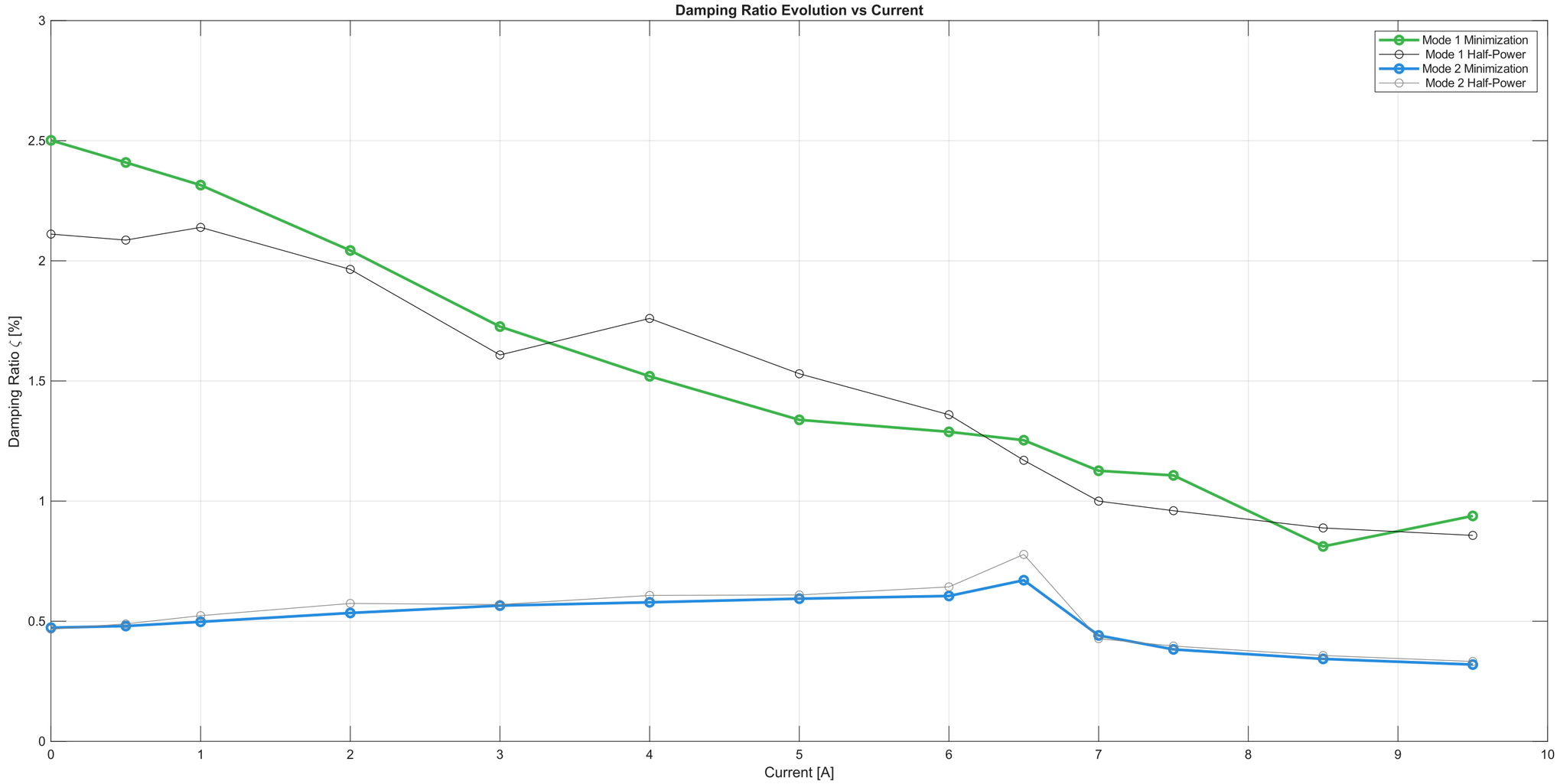
Natural frequency vs. current (Mode 1)



Natural frequency vs. current (Mode 2)



Damping ratio vs Current for Mode 1 and 2



Controller Design

04

Feedforward control Strategy

- **Challenges**

1. **The non-linearity and S-curve phenomenon:** The relationship between the electrical current input (I) and the resulting natural frequency (f) is highly non-linear, taking the shape of a steep S-curve.
2. **The transition zone sensitivity:** The system gain is very high in the phase transformation curve, so a small variation in current results in a strong jump in frequency
3. **Feedback Vulnerability:** Feedback-only loops suffer from severe thermal lag, leading to saturation and limit-cycle chatter during large setpoint steps.

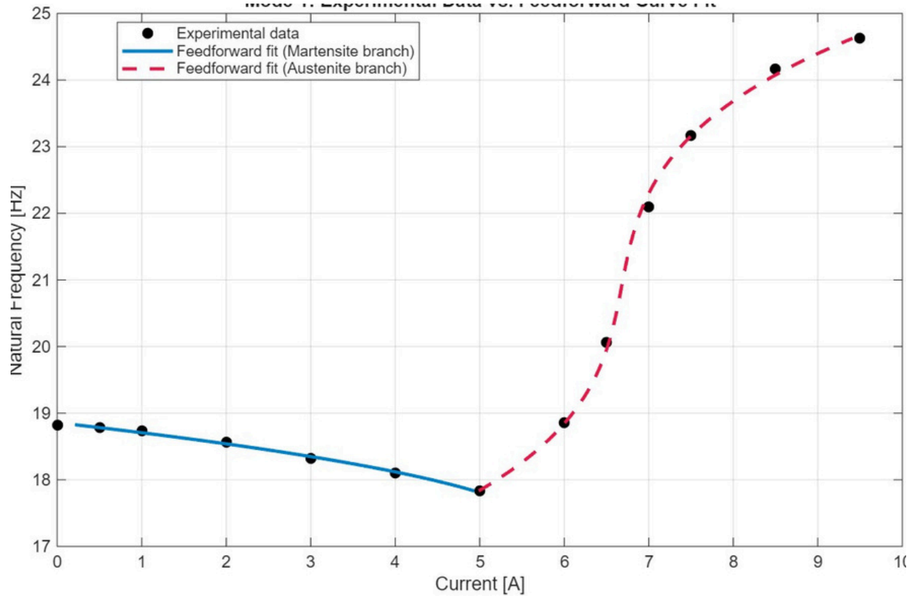
- **Solution**

One of the possible solutions is feedforward control. Unlike feedback control, feedforward is predictive. It measures the dominant frequency in the system and immediately calculates the required current.

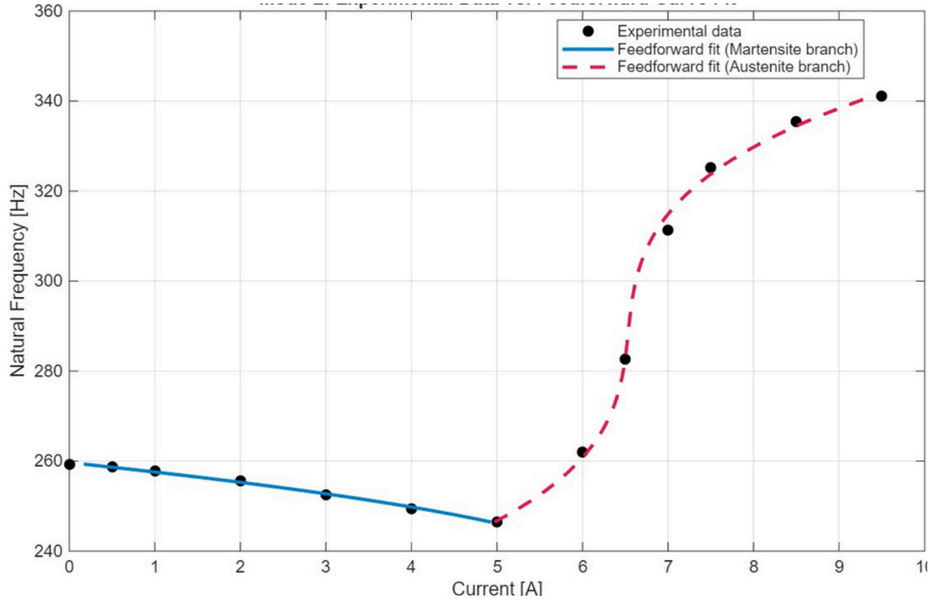
- **Implementation**

Mapping out the steady-state frequency achieved at each current step, and inverting it mathematically gives a polynomial function

Feedforward Control Strategy



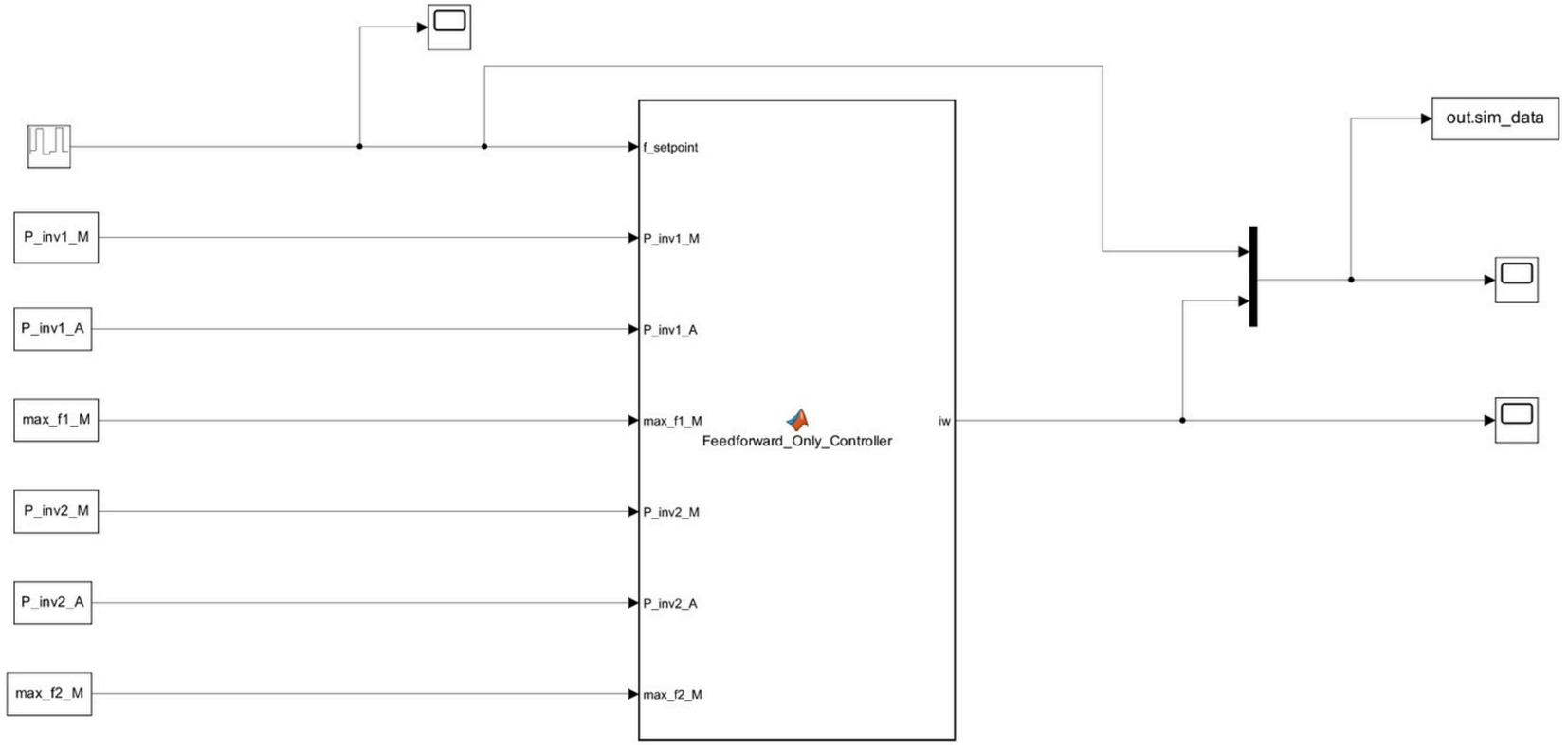
(a) Mode 1: Frequency vs. Current



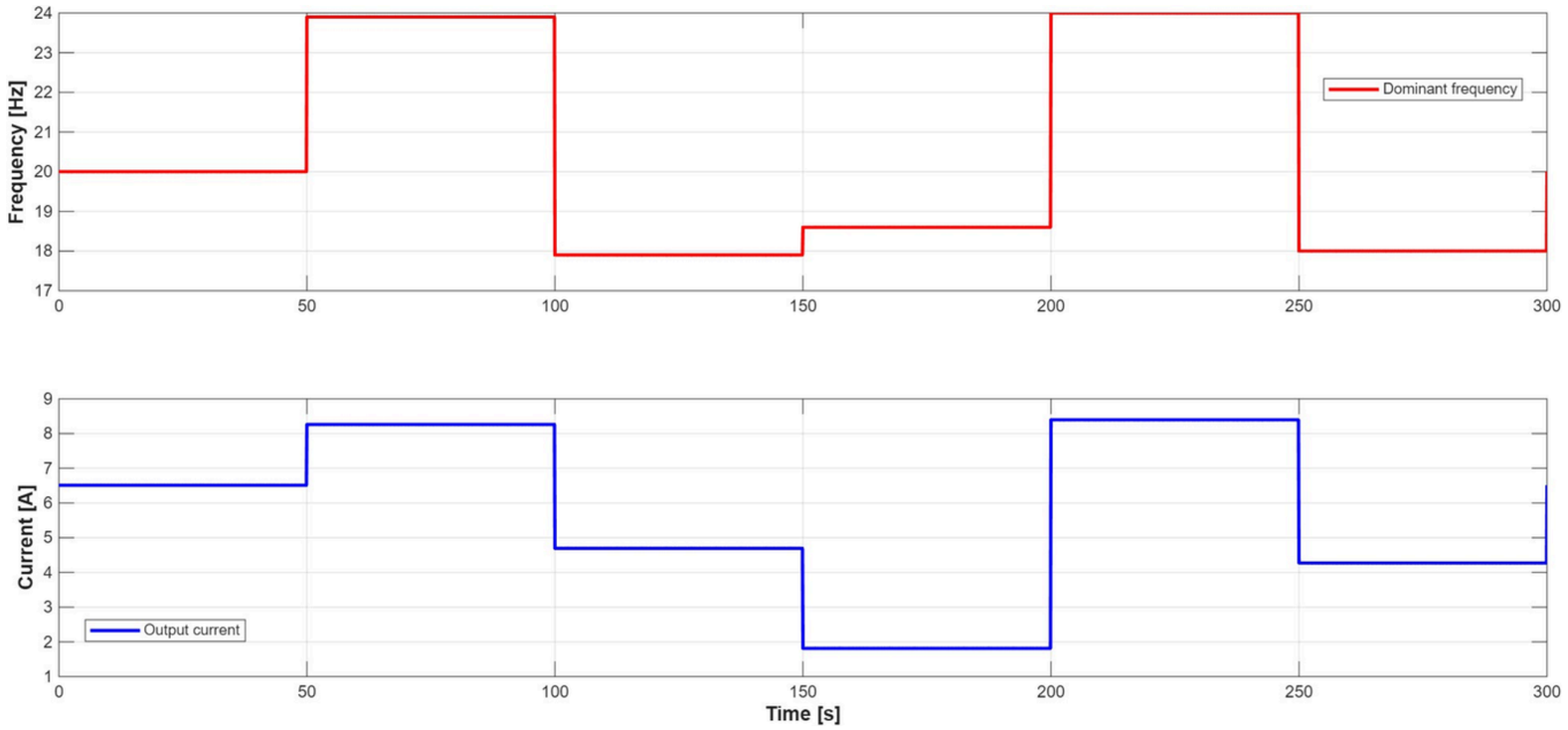
(b) Mode 2: Frequency vs. Current

- **Piecewise Mapping:** Four independent polynomial curves are fitted to capture the highly non-linear phase transformation behavior for both structural modes.
- **Inverse Control:** The system uses these inverse functions to translate the detected dominant frequency setpoint into the precise required control current.

Final Simulink Model of Feedforward Controller

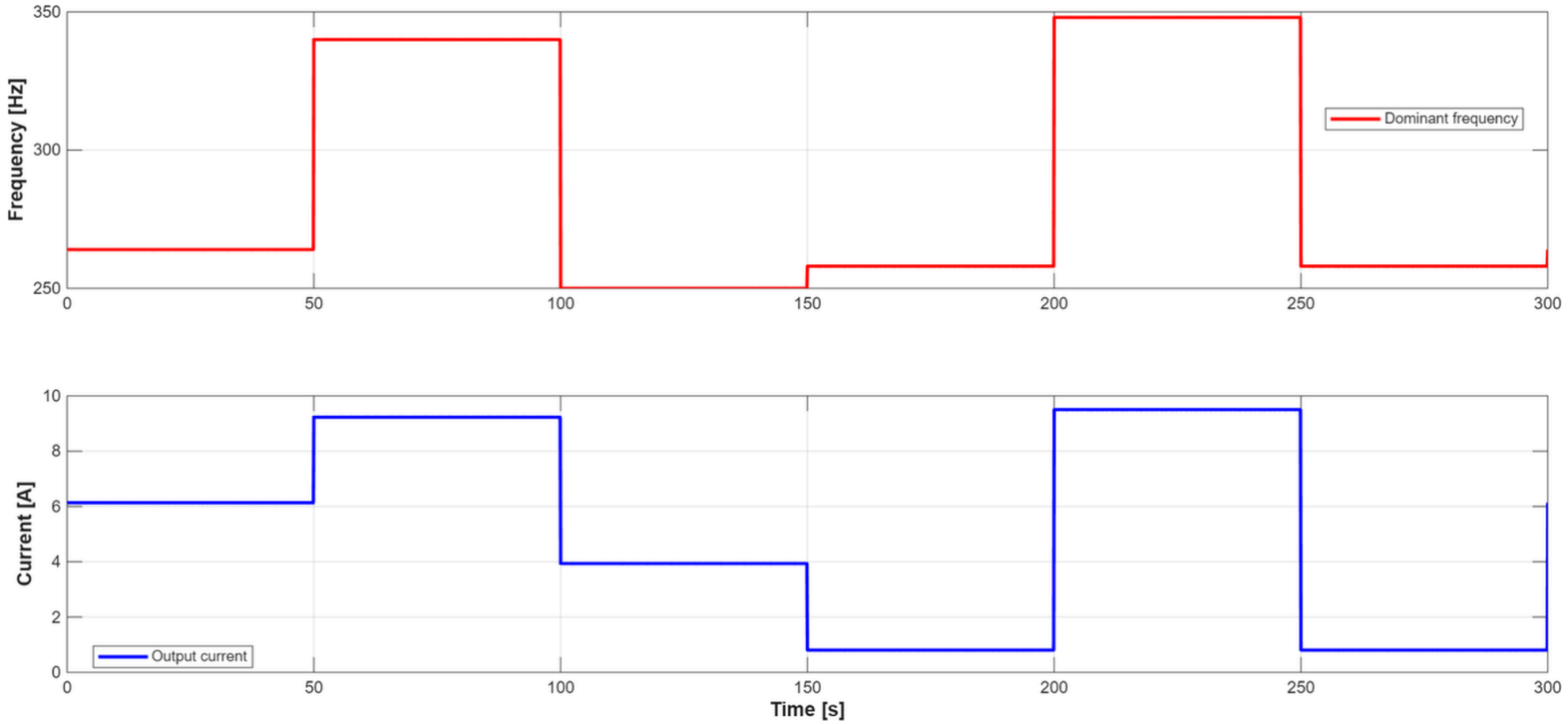


Feedforward Control Result - Mode 1



Instantaneous Response: The controller translates the target frequency into the precise required current with zero computational delay, bypassing the thermal inertia of the SMA.
Dual-Branch Logic in Action: The controller correctly identifies the material's thermal state. For Austenite targets, higher frequencies demand more current. For Martensite targets, the logic correctly reverses, tracking the negative slope of the experimental S-curve.
Deterministic Repeatability: Repeated frequency setpoints yield identical current commands, proving the reliability of the static inverse mapping.

Feedforward Control Result - Mode 2



Instantaneous Response: The controller translates the target frequency into the precise required current with zero computational delay, bypassing the thermal inertia of the SMA.
Dual-Branch Logic in Action: The controller correctly identifies the material's thermal state. For Austenite targets, higher frequencies demand more current. For Martensite targets, the logic correctly reverses, tracking the negative slope of the experimental S-curve.
Deterministic Repeatability: Repeated frequency setpoints yield identical current commands, proving the reliability of the static inverse mapping.

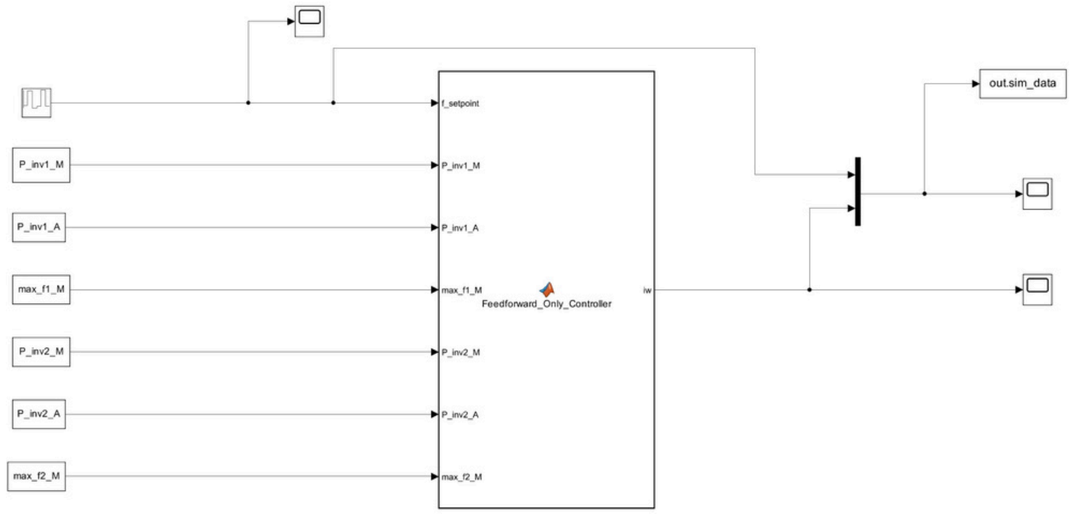
Advantages and limitations of Feedforward Controller

Advantages

- Very simple implementation
- Fast response
- No feedback required

Limitations

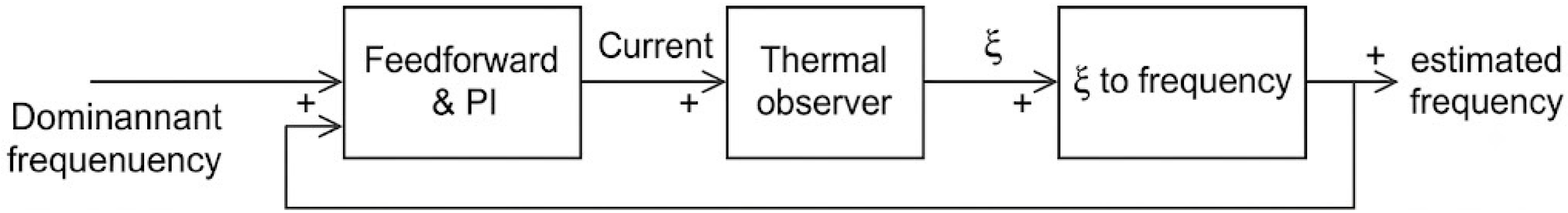
- Sensitive to modelling errors
- Cannot reject disturbances
- No correction for thermal dynamics



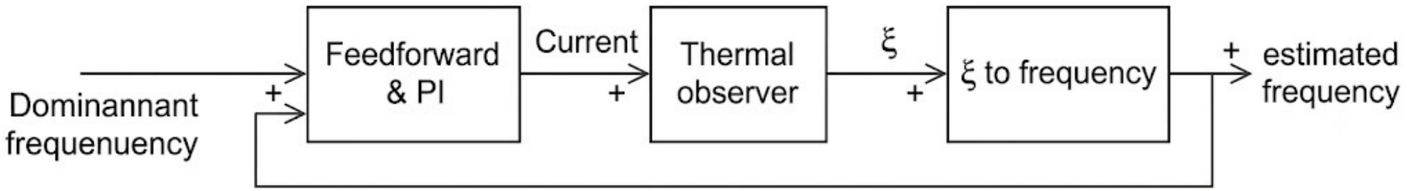
Feedback + Feedforward control

Why is a hybrid approach required?

- **Limitation of Pure Feedback**: Forcing fixed linear gains to traverse the steep, non-linear phase-transition band from zero induces severe limit cycles (chattering) and strong oscillations in the output.
- **The Hybrid Synergy**: Feedforward instantaneously places the system in a stable, linearizable region, while feedback handles only small residual gap correction.



Mathematical formulation of Hybrid controller



1) Feedforward Current Estimation

$$I_{ff} = P^{-1}(f_{target})$$

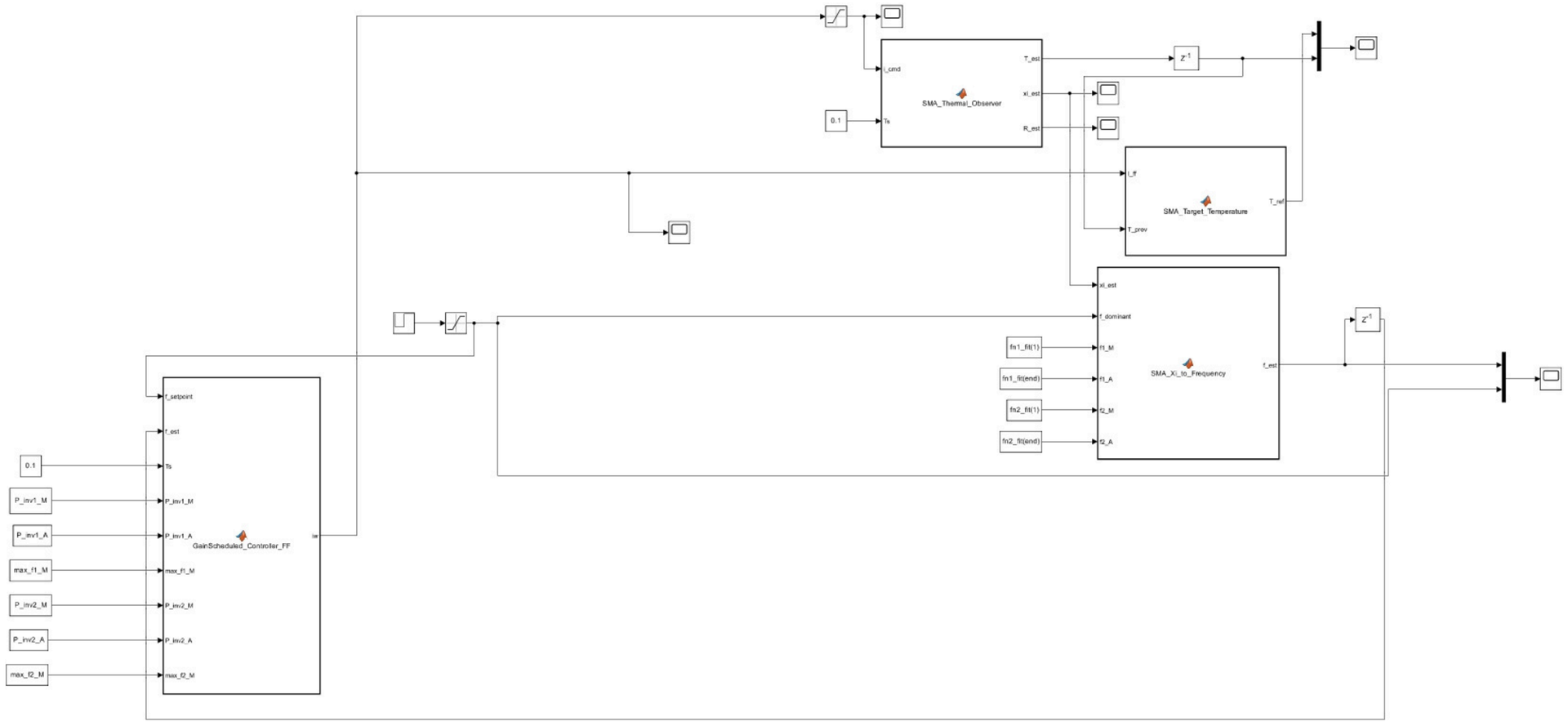
2) Thermal Observer

$$mc_p \frac{dT}{dt} = Ri^2 - hA(T - T_0)$$

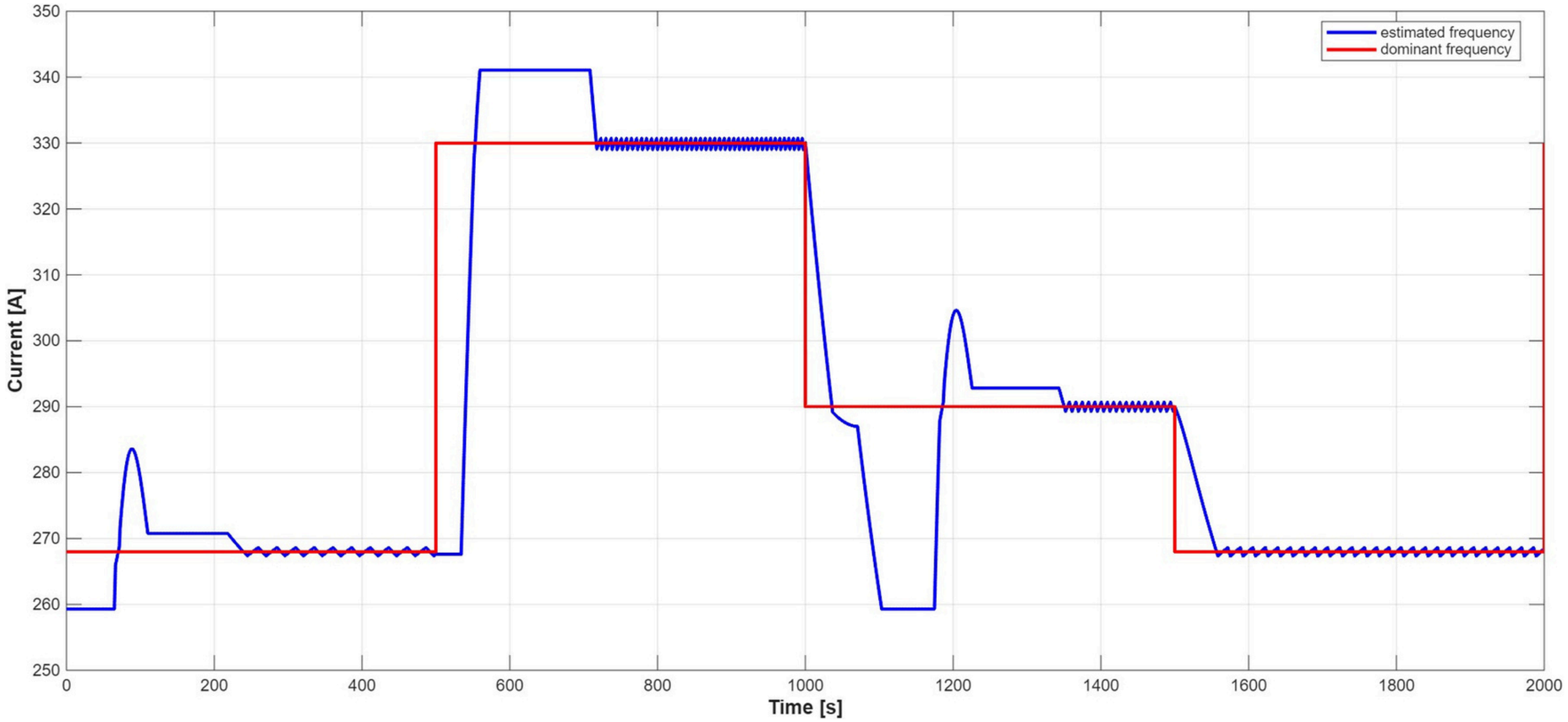
3) Frequency Estimation

$$f = f_M \sqrt{\xi + (1 - \xi)\left(\frac{f_A}{f_M}\right)^2}$$

Final Simulink Model of Hybrid Controller



Result of Hybrid control



Conclusions

05

Summary of results

Tunable stiffness

MODE 1	MODE 2	ADAPTATION
17.8 → 24.6 Hz	247 → 341 Hz	≈ +38%

Extraction Validated

- **Coherence** > 0.95 in 10-500 Hz gives high reliability FRF's (H1 removing clamp mass)
- **SDOF** assumption confirmed by Nyquist circle fit NFQ > 0.999 and RMSE = 0.001-0.02

Controller Design

- Controller is designed and tested in 3 cases
- Advantages and limitations of each one is mentioned
- Final results are explained

Comparison with reference paper

	Reference	Our Work
Phase transition	5-6 A	5-6 A ✓
Austenite reached	7-7.5 A	≈ 7 A ✓
1st Mode (0 A)	18 Hz	18.8 Hz ✓
FRF estimator	H1	H1 ✓
Range tested	< 7.5 A	< 9.5 A

Improvements

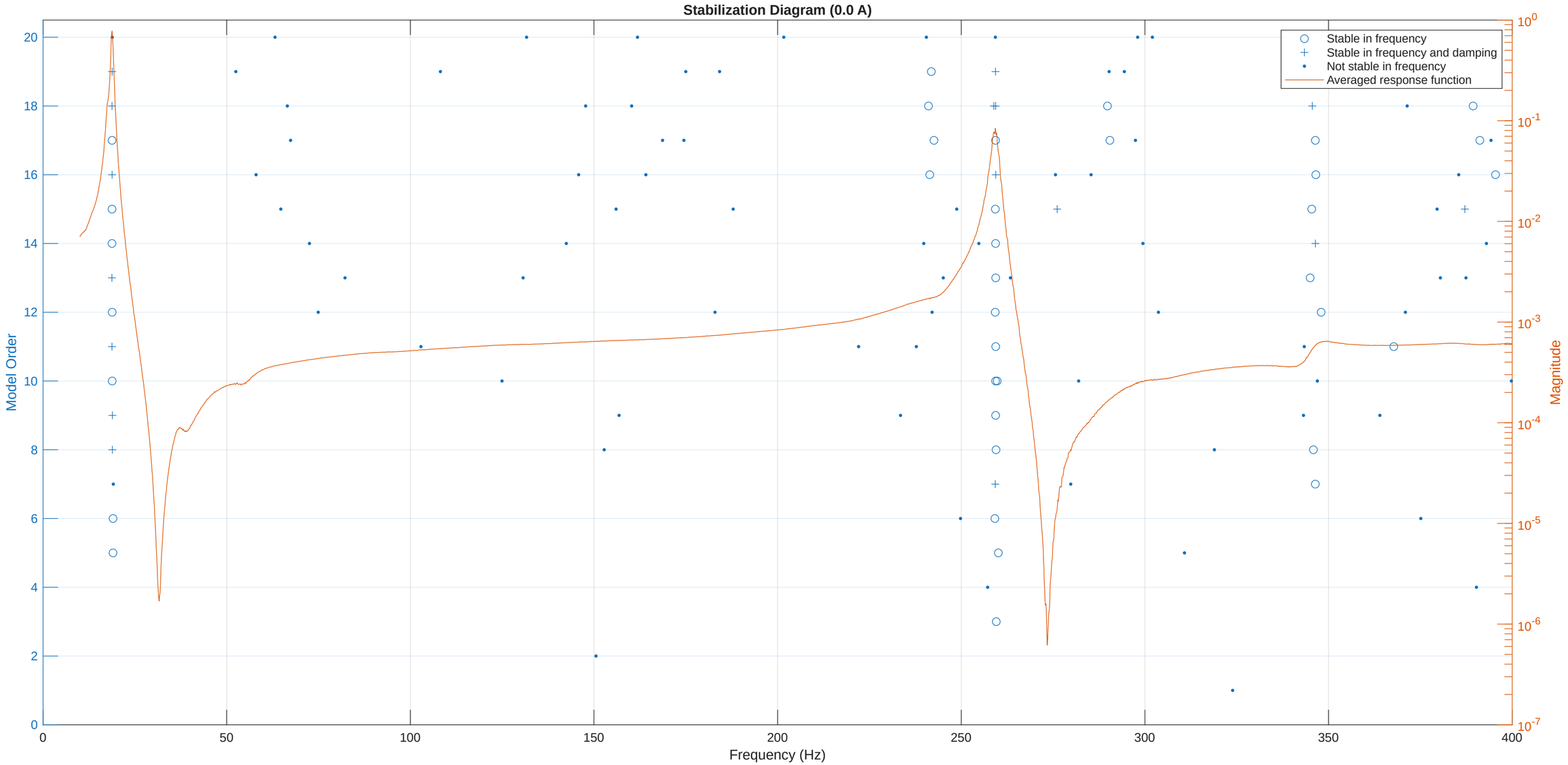
- Cooling: thermal hysteresis not addressed
- Derive **h** coefficient which was assumed constant
- Linear model of a non-linear material
- Improvement of the Theoretical model
- Power-consumption analysis
- Adding measurement points to derive the mode shapes and computation of MAC

Thanks for the attention

Appendix

06

Global MDOF Validation



Global MDOF Validation

